

Several Functions

Look at this code:

```
def double(number):  
    return number * 2  
def square(number):  
    return number * number  
print("1: double\n2: square")  
choice = input("Choice: ")  
value = int(input("Number: "))  
if choice == "1":  
    print("Result:", double(value))  
else:  
    print("Result:", square(value))
```

Use the code above to help you. After completing each challenge below, run your program to test that it works correctly. Save each completed challenge program in a new file.

Programming Challenges

Challenge 1

Change the example's `square()` function into `triple(number)`, which should `return` `number * 3`. Update the menu text and the function call so option "2" triples. For example, "2" and 6 display "Result: 18".

Challenge 2

Extend the example with `half(number)`, which should `return` `number / 2`, and menu option "3: half". Use `if` and `elif` to call the matching function, and `else` to display "Invalid choice." For example, "3" and 5 display "Result: 2.5".

Challenge 3

Create a room calculator with three functions:

`show_menu()` displays "1: area, 2: perimeter", and `area(length, width)` and `perimeter(length, width)` each `return` a result. Call `show_menu()`, ask for a choice, a

length and a width, then call the matching function. Otherwise display "Invalid choice."

